ENGRAM Master Public Technical Report

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All done on a 16GB RAM Mac Mini...in my basement.

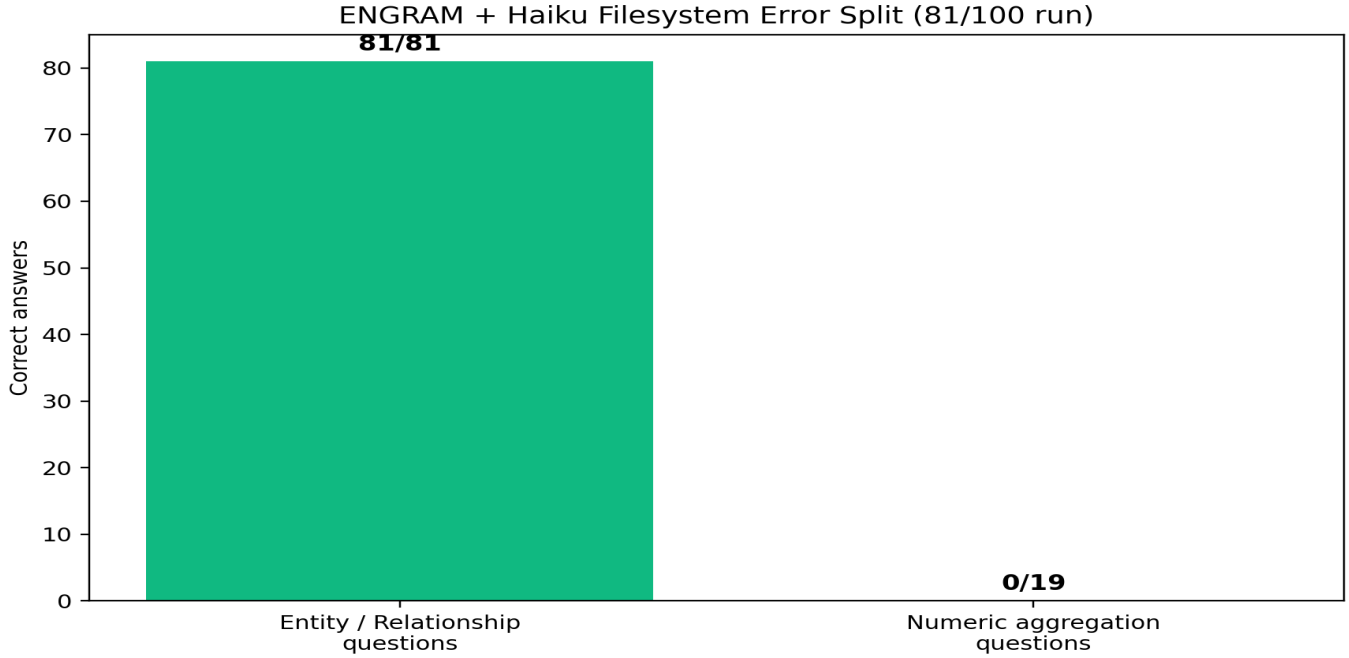
ENGRAM is a memory architecture for LLM agents focused on state-aware memory resolution across long-running contexts.

Internal benchmarks validated architecture. External benchmarks validated against industry/public datasets.

Benchmark	Result
Letta Core Memory READ	96.18% (1058/1100)
Letta Core Memory UPDATE	100.00% (1100/1100)
Letta Filesystem*	81.00% (81/100)
REAL20	98.83% (3953/4000)
RULER	90.15% (1803/2000)
BABILong	88.50% (885/1000)
MultiHop-RAG public	62.56% vs 56.0% published GPT-4

* custom harness + equivalent tool interfaces; not native Letta runtime submission

Filesystem taxonomy: 81/81 entity retrieval, 0/19 numeric aggregation.



Internal benchmark tables

Baseline note: same harness + same Haiku model across retrieval lane baselines.

REAL20 (4000q)

Strategy	Accuracy	Correct/Total	Cost
ENGRAM	98.83%	3953/4000	$9.10
modern_rag	46.95%	1878/4000	$4.84
rag	64.88%	2595/4000	$6.83
vector	65.15%	2606/4000	$7.65
naive	54.60%	2184/4000	$10.90
LlamaIndex(best)†	71.30%	570/800	$0.71

RULER (2000q)

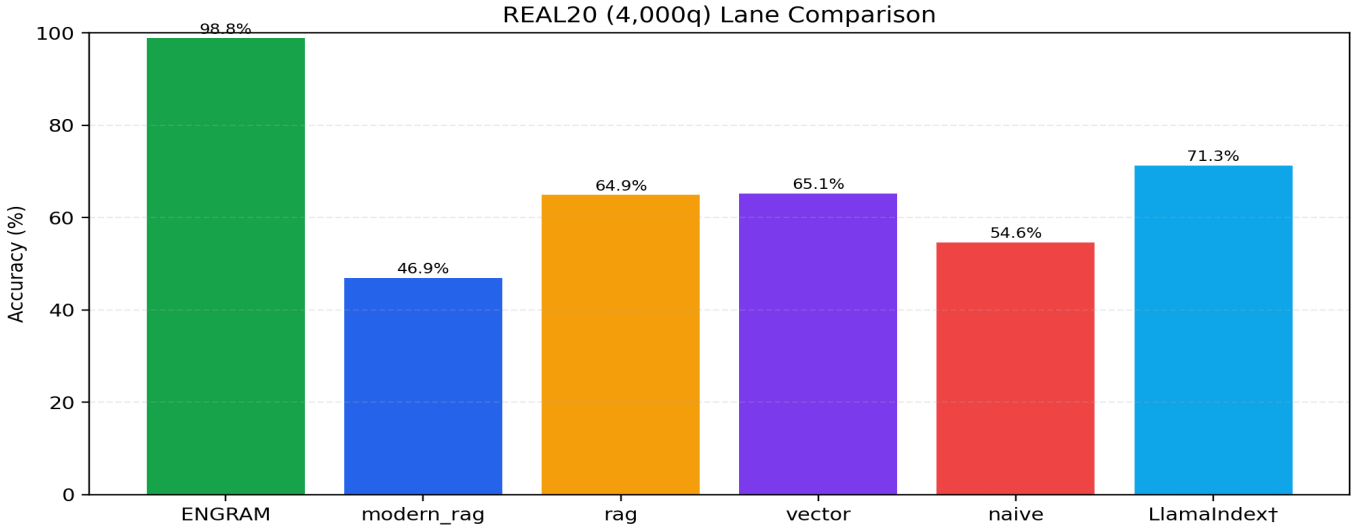
Strategy	Accuracy	Correct/Total	Cost
ENGRAM	90.15%	1803/2000	$2.68
modern_rag	82.40%	1648/2000	$1.98
rag	76.05%	1521/2000	$1.99
vector	76.45%	1529/2000	$2.28
naive	55.00%	1100/2000	$3.15
LlamaIndex(best)†	49.00%	n/a	$0.33

BABILong (1000q)

Strategy	Accuracy	Correct/Total
ENGRAM	88.50%	885/1000
modern_rag	47.30%	473/1000
rag	48.50%	485/1000
vector	49.40%	494/1000
naive	42.90%	429/1000
LlamaIndex(best)†	48.40%	484/1000

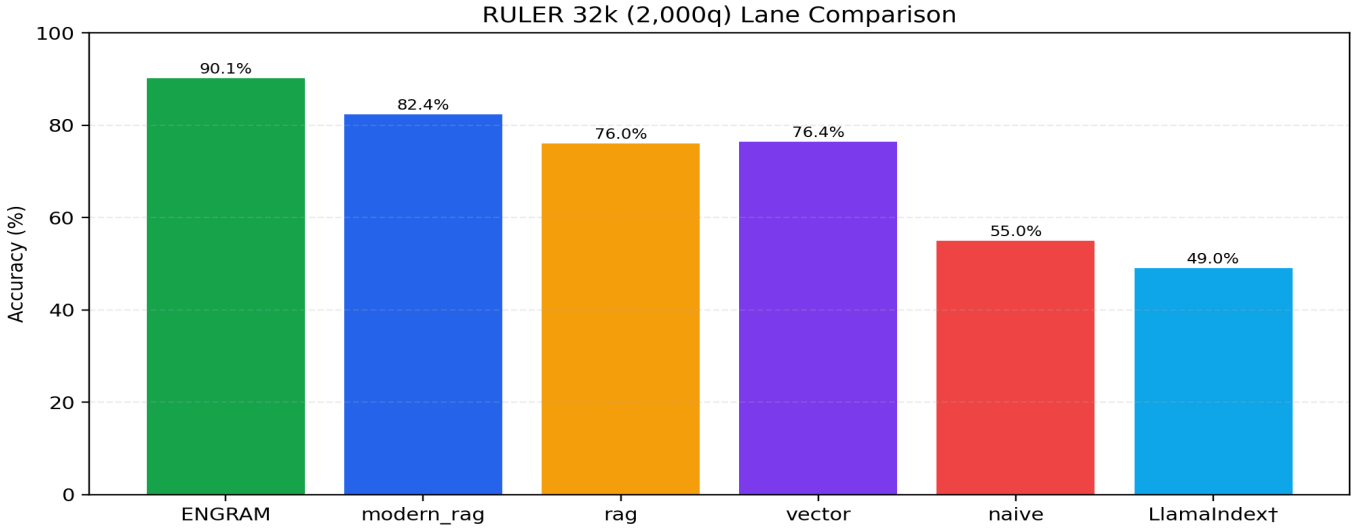
Charts

REAL20 comparison (with LlamaIndex†)



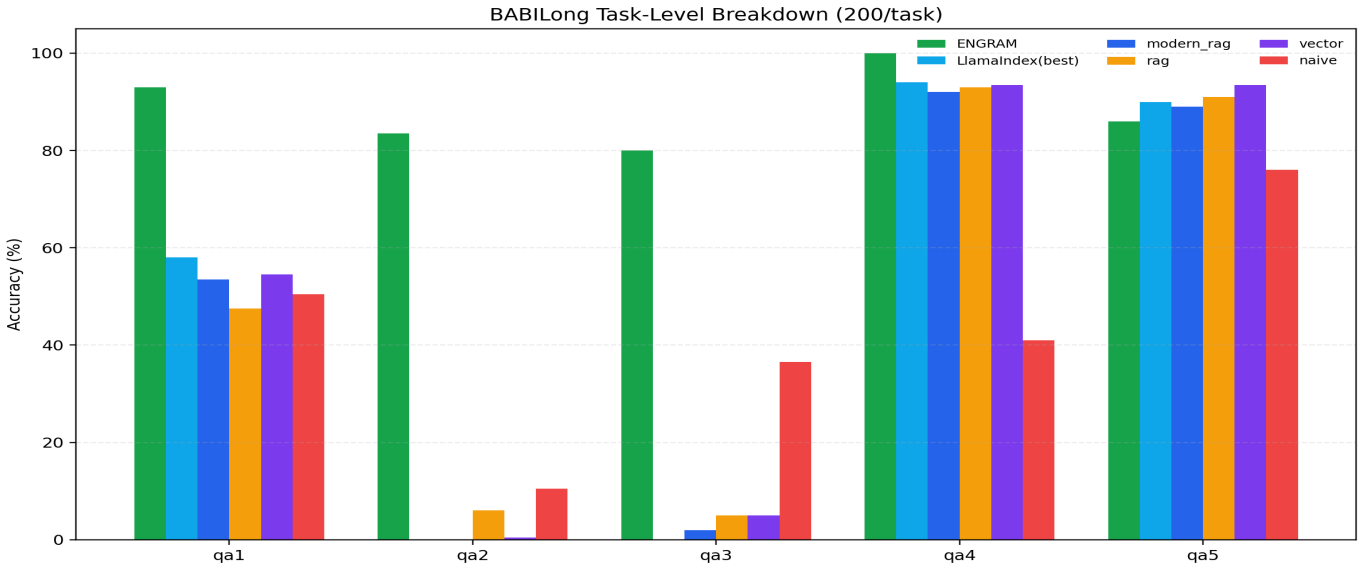
† LlamaIndex from separate best-config sweep (context row, not primary lane harness).

RULER comparison (with LlamaIndex†)

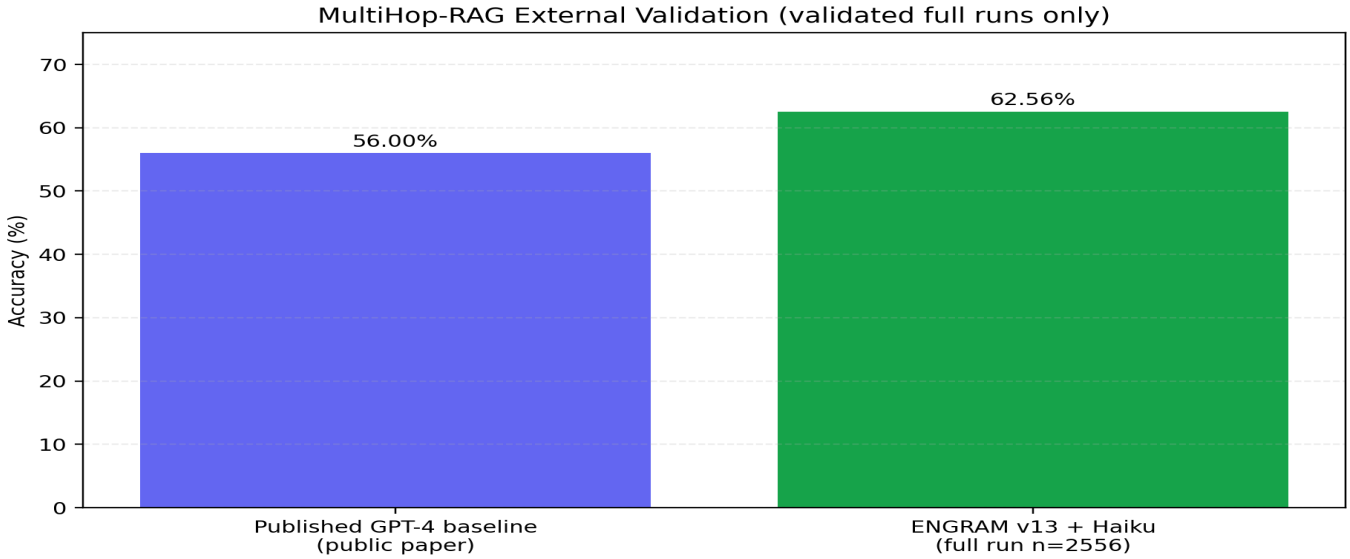


† LlamaIndex from separate best-config sweep (context row, not primary lane harness).

BABILong task-level



MultiHop-RAG validated full-run



Notes

† LlamaIndex values are context rows from separate best-config sweep runs and are labeled consistently in tables/charts.
No fixed Opus dollar claim is included; verify live leaderboard cost snapshot pre-distribution.

Artifacts repo: https://github.com/ENGRAMai-Labs/ENGRAMAI-letta-three-tests-evidence